

Dr. Elena V. Rostova

Research Statement & 5-Year Vision

Candidate for Tenure & Promotion to Associate Professor

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 📄 Code: github.com/rostova-lab
 📅 Evaluation Period: 2021–2026

Tenure Track: 2021–2026 **Grants:** \$2.45M (PI/Lead) **Publications:** 16 Papers **Ph.D. Students:** 4 Grad / 3 Active

1 Overarching Research Vision & Philosophy

My research operates at the intersection of **distributed systems**, **edge computing**, and **fault-tolerant machine learning algorithms**. As autonomous systems—ranging from fleets of robotic vehicles to distributed medical diagnostic sensors—are increasingly deployed in unconstrained physical environments, traditional cloud-centric computing paradigms fail due to high latency, bandwidth limitations, and severe network partitioning.

Core Mission: My laboratory, the *Vertex Distributed Systems Laboratory (VDSL)* at Meridian University, develops the theoretical foundations and system software architectures necessary to realize **provably resilient, real-time autonomous intelligence across heterogeneous, bandwidth-constrained edge networks**.

Over the past five years as an Assistant Professor, my research agenda has centered on three mutually reinforcing pillars:

- 1. Byzantine-Robust Distributed Optimization:** Architecting asynchronous stochastic gradient algorithms capable of maintaining formal convergence guarantees despite arbitrary worker failures, malicious inputs, or extreme network stragglers (Project *Synapse*).
- 2. Heterogeneous Resource-Aware Federated Learning:** Devising dynamic gradient compression and device-scheduling protocols designed for non-IID data distributions on low-power microcontrollers (Project *Nexa*).
- 3. Hardware-Assisted Edge Execution for Swarm Autonomy:** Co-designing real-time control loops and distributed consensus algorithms verified on physical autonomous drone and ground rover testbeds (Project *Apex*).

Since joining Meridian University in Fall 2021, I have established a vibrant, competitive research program funded by **\$2.45M in total competitive external grants** (including an **NSF CAREER Award** in 2024), published **16 peer-reviewed papers** in premier computer science venues (such as *NeurIPS*, *ICML*, *OSDI*, *IEEE TPDS*, and *ACM EuroSys*), graduated **4 Ph.D. students**, and built active interdisciplinary collaborations with national laboratories and industry partners like Synapse Robotics and Nimbus Systems.

2 Major Research Accomplishments During Tenure Track

2.1 Thrust 1: Byzantine-Robust Asynchronous Distributed Learning (Project Synapse)

Distributed machine learning on edge clusters suffers from severe straggler effects and network dropouts. Traditional synchronous parameter server frameworks stall when waiting for slow nodes, while standard asynchronous stochastic gradient descent (SGD) models are vulnerable to poisoned gradients from compromised edge nodes.

Key Breakthroughs in Project Synapse

- **Sub-linear Communication Overhead:** Formulated the *Krum-Asynch* gradient filter, the first asynchronous aggregation rule providing optimal breakdown point ($\frac{1}{2} - \epsilon$) while requiring only $O(d \log K)$ per-step communication complexity for d -dimensional model vectors across K workers.
- **Bounded Asynchrony Under Extreme Drift:** Proved tight convergence bounds under non-convex loss functions when worker staleness follows arbitrary heavy-tailed distributions.
- **Open-Source Impact:** Implemented our algorithms into the *Synapse-ML* framework, currently deployed by over 40 research groups worldwide and integrated into industrial pipelines at Nexa Autonomous Systems.